

Mohammed Ahmer Farooq

North York, ON | mfarooq3@ualberta.ca | 8254193217

Experience

BI & Data Analytics trainee, SLB (Schlumberger)

May 2024 – Dec 2024

- **Sales Performance Dashboarding:** Designed interactive **Power BI** dashboards to track sales performance and operational KPIs, enabling management to monitor regional performance in real-time.
- **Data Collection & Cataloging:** Gathered data from multiple sources; documented and structured datasets into central repositories for **AI/ML** predictive use cases.
- **Predictive Analytics:** Processed raw data into analysis-ready formats, performing exploratory and statistical analyses to support predictive modeling.
- **Agile Methodology:** Worked within an **Agile/Scrum** framework using **JIRA** to manage sprints, automate reporting workflows, and ensure 100% data accuracy in executive dashboards.

Technologies

- **Analytics & BI:** Power BI, Tableau, SQL Server (T-SQL), Excel (Advanced), Statistical Modeling.
- **Data Science:** Python (Pandas, NumPy, Scikit-learn), R (Aggregation), Feature Selection, Parameter Tuning.
- **Automation & Tools:** Process Automation, JIRA, Agile (Scrum/Kanban), Git/GitHub, Docker.
- **Data Engineering:** ETL/ELT Pipelines, Data Cleaning & Validation, Transactional Data Structures.

Projects

Quantitative Market Dynamics & Predictive Modeling

- **High-Dimensional Feature Engineering:** Synthesized a library of 50+ technical indicators, including exponential moving averages, RSI, and Fourier Transforms, to capture cyclicity and momentum in non-stationary time-series data.
- **Ensemble Architecture:** Developed a robust stacking ensemble utilizing XGBoost, LightGBM, and Random Forest; optimized hyperparameters via Bayesian Search to improve directional accuracy by 15% over baseline models.
- **Quantitative Backtesting Engine:** Engineered a rigorous walk-forward validation framework to simulate real-world trading constraints, incorporating transaction costs and slippage to mitigate look-ahead bias and overfitting.

StudyFlow – AI Orchestration System

- **AI Dashboard Integration:** Built a React-based dashboard featuring an AI agent that automatically parses unstructured data (PDFs/Notes) into a connected "Knowledge Graph."
- **Intelligent Retrieval:** Implemented a RAG-based system using Python and LLMs to automate link-finding across complex datasets, enabling "chat-with-data" functionality.
- **Automated Data Validation:** Engineered a data-cleaning layer to filter noise from raw input, ensuring 100% accuracy in AI-generated summaries and visualizations.

Education

University of Alberta, BS in Computer Science

Sept 2022 – January 2027

- Coursework: Machine Learning, Comparison of Learning Algorithms, Web architecture

Certifications

Databricks Fundamentals Accreditation, Databricks

Machine Learning Foundations, Amii